

Species-Aware Module Review: 2-Methylcitrate Cycle in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI 160488; proteome UP000000556) **Bucket:** KEGG ppu00640 "Propanoate metabolism" **Scope:** Propionate → propionyl-CoA → 2-methylcitrate → 2-methylisocitrate → succinate + pyruvate **Date:** 2026-07-23 · Finalized across Iterations 1–3 **Evidence base:** UniProt/InterPro + KEGG KO cross-check + 7 literature references. No KT2440-specific enzymology exists; all conclusions are orthology/annotation-based (see §5, Appendix A).

1. Executive Summary

The 2-methylcitrate cycle (2-MCC) is **fully satisfiable** in *P. putida* KT2440. All five module steps are encoded by high-confidence candidates concentrated in a single chromosomal cluster, **PP_2334–PP_2339** (*prpB–prpC–acnA-II–prpF–prpD–acnB*), plus **PP_2351** (*prpE*, propionyl-CoA synthetase). The 2-MCC is the *only* characterized propionate-oxidation route in *Pseudomonas* (

P 37995793

), so the module should be marked **covered**.

The most curation-relevant finding is that KT2440 encodes **both** 2-methylcitrate-dehydration implementations that the generic module treats as mutually exclusive alternatives: - the **direct, cofactor-less PrpD route** (PP_2338), and - the **two-component Fe/S AcnD/PrpF route** (PP_2336 + PP_2337).

PP_2336 (labelled "acnA-II") is not a generic aconitase — it carries the diagnostic InterPro **IPR012708** (Fe/S-dependent 2-methyl-isocitrate/2-methylcitrate dehydratase, the AcnD family) and sits immediately beside a bona fide **PrpF** gene (IPR012709 / PF04303), which is required only in AcnD-using organisms. This resolves the module's "AcnA-II/AcnD candidate" to **AcnD (covered)**.

Main curation actions: reinterpret PP_2336 as AcnD; add EC 2.3.3.5 / methylcitrate-synthase GO to PP_2335 (currently the over-broad "Citrate synthase, no EC"); treat PrpE substrate specificity as context-inferred; note all cluster entries are UniProtKB TrEMBL (**unreviewed**).

2. Target-Organism Pathway Definition

Included biochemistry (5 steps): 1. Propionate activation to **propionyl-CoA** — PrpE (ATP-dependent, AMP-forming ligase). 2. Condensation of propionyl-CoA + oxaloacetate → **(2S,3S)-2-methylcitrate** — PrpC (2-methylcitrate synthase). 3. Dehydration of 2-methylcitrate → **2-methyl-cis-aconitate** — PrpD (direct) **or** AcnD+PrpF (two-component). 4. Hydration of 2-methyl-cis-aconitate → **2-methyl-D-threo-isocitrate** — aconitase-family AcnB. 5. Aldol cleavage of 2-methylisocitrate → **succinate + pyruvate** — PrpB (2-methylisocitrate lyase).

Neighboring maps to keep separate (do NOT merge into this module): - TCA-cycle aconitase (*acnA-I* PP_2112, *acnB* PP_2339 housekeeping role) and citrate synthase (*gltA*) — ppu00020. - Methylmalonyl-CoA / vitamin-B12 propionate route (*mmsA*, methylmalonate-semialdehyde; ppu00562/00640 fringe) — a *distinct* propionate pathway not used here. - Fatty-acid β -oxidation and BCAA/BCKA dehydrogenase genes (*fadB*, *paaF*, *bkd*, *lpd*, *acd*, *acsA*, *acc*, *pta*) — these are ppu00061/00071/00280/00785 members that appear in the candidate list only because ppu00640 is a broad KEGG "overview" bucket; they are **not** 2-MCC steps.

Alternate names / DB definitions: "2-methylcitric acid cycle (2-MCC)", "methylcitrate cycle (MCC)", "propionate/propionyl-CoA oxidation pathway". KEGG folds it inside map00640 (Propanoate metabolism) with module **M00741** (propanoyl-CoA → succinyl-CoA is the *other* route, not this one) — the 2-MCC proper corresponds to the *prpBCDE* gene set, not a single tidy KEGG module.

3. Expected Step Model vs. PSEPK Genes

#	Module step	Enzyme / EC	PSEPK gene	Status
1	Propionate activation	Propionyl-CoA synthetase, 6.2.1.17	prpE PP_2351 (Q88KD9)	covered (specificity inferred)
2	2-methylcitrate synthesis	2-methylcitrate synthase, 2.3.3.5	prpC PP_2335 (Q88KF5)	covered (EC missing)
3a	Dehydration (direct)	2-MC dehydratase, 4.2.1.79	prpD PP_2338 (Q88KF2)	covered
3b	Dehydration (2-component)	AcnD Fe/S dehydratase + PrpF isomerase	acnA-II/AcnD PP_2336 (Q88KF4) + prpF PP_2337 (Q88KF3)	covered
4	Hydration to 2-methylisocitrate	Aconitase-family, 4.2.1.99	acnB PP_2339 (Q88KF1); paralog acnA-I PP_2112 (K27802)	covered (redundant paralogs)
5	C-C cleavage	2-methylisocitrate lyase, 4.1.3.30	prpB PP_2334 (Q88KF6)	covered

No step is a **gap** or **not_expected_in_target_taxon**.

4. Candidate Genes and Evidence

High-confidence, in-cluster (promote-worthy):

- **PrpE — PP_2351 (Q88KD9, 629 aa).** AMP-binding acyl-CoA ligase (Pfam PF00501 + ACAS_N PF16177). Role: propionate → propionyl-CoA. Caveat: the AMP-binding family is broad; propionyl- vs acetyl-CoA specificity is inferred from *prp*-cluster context (KT2440 also has dedicated acetyl-CoA synthetases *acsA1/acsA2*), not from a PSEPK biochemical assay. *Salmonella* PrpE is a propionyl-CoA synthetase

(P 10482501).

- **PrpC — PP_2335 (Q88KF5, 375 aa).** InterPro **IPR011278** "2-MeCitrate/Citrate_synth_II" = the type-II (methyl)citrate synthase clade. Function is well established across bacteria; PrpC prefers propionyl-CoA

(
P 10482501
);

C. glutamicum PrpC has 2-methylcitrate synthase activity,

P 11976302
).

Caveat: UniProt calls it "Citrate synthase" with **no EC** — over-broad; should be EC 2.3.3.5.

- **PrpD — PP_2338 (Q88KF2, EC 4.2.1.79).** Cofactor-less 2-methylcitrate dehydratase. Direct-route enzyme.

- **AcnD — PP_2336 (Q88KF4, 862 aa).** Carries diagnostic **IPR012708** (Fe/S 2-methylcitrate/2-methylisocitrate dehydratase = AcnD) with 4Fe-4S/Iron-sulfur keywords. AcnD performs only the first half-reaction (dehydration to a 4-methylaconitate isomer) and requires PrpF

(
P 14702315
);

P 29145506
).

Caveat: UniProt annotates it merely as "aconitate hydratase (EC 4.2.1.3)"; reinterpret as AcnD.

- **PrpF — PP_2337 (Q88KF3, 396 aa).** InterPro **IPR012709** / Pfam **PF04303** "PrpF"; isomerase. Isomerizes the AcnD product into 2-methyl-*cis*-aconitate, the substrate of housekeeping aconitase

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P 17567742
);

P 29145506
).

Presence of PrpF is itself strong evidence the AcnD route is functional.

- **AcnB — PP_2339 (Q88KF1, 869 aa, EC 4.2.1.3/4.2.1.99).** Aconitase-B family (IPR004406); catalyzes the hydration of 2-methyl-*cis*-aconitate to 2-methylisocitrate (the module's aconitase-family step).
- **PrpB — PP_2334 (Q88KF6, 296 aa).** InterPro **IPR012695** "PrpB"; 2-methylisocitrate lyase (EC 4.1.3.30), Mg-dependent. Cleaves 2-methylisocitrate → succinate + pyruvate. A *Pseudomonas prpB* mutant is blocked in propionate oxidation

(
P 37995793
).

Candidate-list members that are NOT 2-MCC steps (over-inclusion by the broad bucket): accA/B/C/D (fatty-acid/malonyl-CoA, ppu00061), pta, PP_2213/acsA1/acsA2 (acetate activation), acnA-I PP_2112 (housekeeping TCA aconitase), fadB/paaF/PP_2217/acd (β -oxidation/CoA hydratases), mmsA-I/II & PP_0596 (methylmalonate-semialdehyde route), sucC/sucD (succinyl-CoA ligase), and the bkd/lpd/lpdG BCKA-dehydrogenase complex (ppu00785). These should stay mapped to their own modules.

5. Gaps, Ambiguities, and Likely Over-Annotations

- **No true gaps.** Every step has a diagnostic-family candidate.
- **Redundant dehydration:** PrpD and AcnD/PrpF coexist — the generic module's "alternative implementations" are **both realized**, not either/or. Module boundary text should allow parallel co-encoded implementations in a single genome.
- **Over-broad annotations to fix:** PrpC "Citrate synthase (no EC)" → methylcitrate synthase (2.3.3.5); AcnD PP_2336 "aconitate hydratase (4.2.1.3)" → AcnD Fe/S 2-methylcitrate dehydratase.
- **Paralog ambiguity (aconitases):** three aconitase-family proteins (AcnA-I PP_2112, AcnD PP_2336, AcnB PP_2339) share EC 4.2.1.3 by homology. AcnB does the 2-MCC hydration; KEGG also credits AcnA-I (K27802) with 2-methylisocitrate-dehydratase activity, so AcnA-I is *not* purely housekeeping. AcnD performs only the dehydration half-reaction. Only the "2-methyl-" activities relevant to the 2-MCC should be curated on the correct paralogs.
- **Paralog ambiguity (lyase superfamily):** PrpB (PP_2334, 2-methylisocitrate lyase, EC 4.1.3.30) and AceA (PP_4116, isocitrate lyase, EC 4.1.3.1) share the ICL/PEP-mutase fold (IPR039556) but have distinct substrate pockets (Salmonella PrpB structure,

P 14575713

AceA is a **decoy** in the ppu00640 bucket — it belongs to the glyoxylate shunt, not the 2-MCC. Map the cleavage step to PrpB only and exclude AceA to prevent over-propagation of 2-methylisocitrate-lyase activity.

- **Evidence level:** all findings rest on *homology + genomic context + strong ortholog biochemistry* in Salmonella/Shewanella/Vibrio/Corynebacterium. There is **no KT2440-specific enzymology** in the retrieved literature; all UniProt entries are TrEMBL (unreviewed, electronic).

6. Module and GO-Curation Recommendations

Module step	Recommended mark
Propionate activation (PrpE)	covered (note: specificity inferred)
2-methylcitrate synthesis (PrpC)	covered (request EC 2.3.3.5 / GO:0050440 addition)
Direct PrpD dehydration	covered
AcnD/PrpF two-component dehydration	covered (reassign PP_2336 → AcnD)
Aconitase-family hydration (AcnB)	covered
2-methylisocitrate cleavage (PrpB)	covered

- **Module needs minor revision:** allow the two dehydration implementations to be *simultaneously* present (parallel), not modeled as exclusive alternatives, since *Pseudomonas* carries both.
- **GO requests:** ensure GO:0050440 (2-methylcitrate synthase), GO:0047547 (2-methylcitrate dehydratase), GO:0046421 (methylisocitrate lyase), and an AcnD/2-methyl-cis-aconitate isomerase (PrpF, GO:0016853 isomerase / trans-aconitate isomerase) term are attached to the right loci.
- No new module document is required; the generic outline already captures both routes.

7. Genes to Promote to Full fetch-gene Review

1. **PP_2336 (acnA-II/AcnD)** — highest priority: annotation says "aconitase" but InterPro says AcnD; needs functional reassignment.
2. **PP_2335 (prpC)** — add methylcitrate-synthase EC/GO; distinguish from housekeeping gltA.
3. **PP_2351 (prpE)** — confirm propionyl-CoA (vs acetyl-CoA) specificity.
4. **PP_2337 (prpF)** — confirm isomerase EC/GO (currently no EC).
5. **PP_2339 (acnB)** — confirm 2-methylisocitrate-dehydratase (4.2.1.99) role vs housekeeping aconitase.

8. Key References

- Horswill & Escalante-Semerena 1999, *Salmonella typhimurium* 2-MCC
([P 10482501](#)).
- Grimek & Escalante-Semerena 2004, AcnD is a new Fe/S 2-methylcitrate dehydratase requiring PrpF
([P 14702315](#)).
- Garvey et al. 2007, PrpF crystal structure / aconitate isomerase
([P 17567742](#)).
- Rocco et al. 2017, PrpF isomerizes the AcnD product to the AcnB substrate
([P 29145506](#)).
- Claes et al. 2002, *C. glutamicum* prpDBC / 2-methylcitrate synthase
([P 11976302](#)).
- Simanshu et al. 2003, *Salmonella* PrpB (2-methylisocitrate lyase) crystal structure; PrpB vs isocitrate-lyase substrate specificity
([P 14575713](#)).
- Santos-Oliveira et al. 2024, *Pseudomonas* prpB mutant; 2-MCC is the only propionate-oxidation route
([P 37995793](#)).

Uncertainty statement: Conclusions are strong by orthology + diagnostic InterPro families + genomic clustering, but no direct *P. putida* KT2440 enzymology was found in the retrieved literature; all UniProt annotations are unreviewed (TrEMBL). Species transfer from *Salmonella*/*Shewanella*/*Vibrio* to PSEPK is **strong** for pathway logic and **moderate** for individual substrate specificities.

Appendix A — Independent KEGG KO cross-check (added Iteration 2)

An orthology system independent of UniProt/InterPro (KEGG KO) agrees on every step and adds detail:

Locus	Gene	KEGG KO	KO name / EC	Interpretation
PP_2334	prpB	K03417	methylisocitrate lyase, EC 4.1.3.30	confirms PrpB
PP_2335	prpC	K01659	2-methylcitrate synthase, EC 2.3.3.5	supplies the EC missing from UniProt
PP_2336	acnA-II	K20455	2-methylcitrate dehydratase (2-methyl-<i>trans</i>-aconitate forming), EC 4.2.1.117	confirms AcnD (distinct KO from PrpD)
PP_2337	prpF	K09788	2-methylaconitate isomerase, EC 5.3.3.-	confirms PrpF isomerase
PP_2338	prpD	K01720	2-methylcitrate dehydratase, EC 4.2.1.79	confirms PrpD
PP_2339	acnB	K01682	aconitate hydratase 2 / 2-methylisocitrate dehydratase, EC 4.2.1.3/4.2.1.99	confirms AcnB (hydration)
PP_2351	prpE	K01908	propionyl-CoA synthetase, EC 6.2.1.17	confirms PrpE specificity (reduces earlier caveat)
PP_2112	acnA-I	K27802	aconitate hydratase A / 2-methylisocitrate dehydratase, EC 4.2.1.3/4.2.1.99	AcnA-I also carries the hydration activity → step-4 redundancy

Consequences for curation: 1. **AcnD confirmed twice.** PP_2336 is AcnD by KEGG KO **K20455 / EC 4.2.1.117** and by InterPro IPR012708. Its product is 2-methyl-*trans*-aconitate, which PrpF (2-methylaconitate isomerase, EC 5.3.3.-) converts to the 2-methyl-*cis*-aconitate accepted by aconitase — this fully resolves the module's "AcnD-family dehydration → PrpF isomerization" sub-steps to real genes with matched product chemistry. 2. **PrpC EC and PrpE specificity are corroborated** by KEGG (EC 2.3.3.5; EC 6.2.1.17), so those caveats downgrade from "over-broad"

to "UniProt EC field incomplete — fill from KEGG." 3. **Step 4 (hydration) is redundant:** both **AcnB (PP_2339)** and **AcnA-I (PP_2112)** carry 2-methylisocitrate-dehydratase capability (EC 4.2.1.99). Mark the step covered with a paralog note; do not attribute it solely to AcnB, and conversely do not treat AcnA-I as purely housekeeping. 4. **No in-cluster regulator:** PP_2340 (between *acnB* and *prpE*) is K06938 "uncharacterized"; no *prpR*-type regulator was found immediately adjacent (regulation likely encoded elsewhere — an open question).

Boundary confirmation: KEGG's ppu00640 gene set is identical to the 33-gene candidate list, confirming that list is the *broad overview map* (Propanoate metabolism), not the 2-MCC module proper. Only PP_2334–PP_2339 + PP_2351 (+ PP_2112 as a hydration paralog) belong to the 2-MCC; the remaining ~25 genes stay in their own modules.

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